Division B Water Quality Test

Milpitas Invitational



Names: _____

Team Name: _____ Team #: _____

Score: _____/100

Test written by Natasha Tiwari and Anvi Gadkari

MCQ (1 point each):

1. Why does low dissolved oxygen often cause fish kills?
    A) Fish overheat
    B) Fish cannot breathe without oxygen in the water
    C) It makes water too clear
    D) Algae grow too fast

2. Which organisms are at the base of most aquatic food chains?
    A) Fish
    B) Algae and phytoplankton
    C) Zooplankton
    D) Decomposers

3. The process by which water evaporates, condenses, and returns as precipitation is called the:
    A) Carbon cycle
    B) Nitrogen cycle
    C) Water cycle
    D) Oxygen cycle

4. A high BOD usually means:
    A) Water is clean and has lots of oxygen
    B) Water is polluted with organic waste
    C) There is low bacterial activity
    D) There are few nutrients

5. If the pH of a lake changes from 7 to 5, the water has become:
    A) More basic
    B) More neutral
    C) More acidic
    D) Cleaner

6. Fecal coliform levels are used to measure:
    A) Temperature of water
    B) Bacteria contamination from animal waste
    C) The number of algae in water
    D) Salt content

7. A sudden increase in algae followed by a decrease in fish is an example of:
    A) Eutrophication
    B) Photosynthesis
    C) Sedimentation
    D) Filtration

8. What role do decomposers play in aquatic food webs?
    A) Eat large fish
    B) Break down dead plants and animals
    C) Produce oxygen
    D) Filter pollutants

9. Biochemical Oxygen Demand (BOD) measures:
    A) How much oxygen bacteria need to decompose organic matter
    B) How much oxygen fish use when swimming
    C) The number of bacteria in the water
    D) The temperature of the water

10. When temperature increases, dissolved oxygen usually:
    A) Increases
    B) Decreases
    C) Stays the same
    D) Doubles

11. Which two parameters are most directly related to each other?
    A) Turbidity and pH
    B) Temperature and dissolved oxygen
    C) Phosphate and salinity
    D) Fecal coliform and total solids

12. What nutrient causes algal blooms when present in excess?
    A) Iron
    B) Nitrates and phosphates
    C) Chlorine
    D) Calcium

13. A watershed is:
    A) A storage tank for water
    B) The area of land that drains into a body of water
    C) Another name for a river
    D) A water treatment plant

14. Which of the following increases turbidity?
    A) Algal growth
    B) Sediment runoff
    C) Erosion
    D) All of the above

15. Plants take in nitrates mainly to:
    A) Build proteins and DNA
    B) Make sugar
    C) Absorb water
    D) Produce oxygen

16. When sediment covers fish eggs or blocks sunlight for plants, this is called:
    A) Oxygenation
    B) Sedimentation pollution
    C) Acidification
    D) Runoff reduction

17. Total solids include:
    A) Only dissolved minerals
    B) Suspended and dissolved particles
    C) Only organic matter
    D) None of the above

18. Dissolved oxygen (DO) is important because:
    A) Fish need it to breathe
    B) It keeps water warm
    C) It makes water taste salty
    D) It increases acidity

19. Harmful invasive species often affect ecosystems by:
    A) Increasing biodiversity
    B) Outcompeting native species
    C) Helping clean water
    D) Fixing nitrogen

20. High turbidity means:
    A) The water is very clear
    B) There are many suspended particles blocking light
    C) There is no sediment in the water
    D) The water has a high pH

21. Which of the following best describes salinity?
    A) The amount of oxygen in water
    B) The amount of salt dissolved in water
    C) The number of bacteria in water
    D) The temperature of the water

22. Why is sedimentation a problem in watersheds?
    A) It increases oxygen for fish
    B) It clogs habitats and reduces light
    C) It removes nutrients
    D) It increases pH

23. The purpose of a primary treatment in wastewater plants is to:
    A) Kill bacteria
    B) Remove large solids and debris
    C) Add chlorine
    D) Filter dissolved minerals

24. Secondary wastewater treatment mainly removes:
    A) Nutrients and bacteria
    B) Metals and rocks
    C) Sand and gravel
    D) Salt

25. What does pH measure?
    A) Acidity or basicity of water
    B) Amount of salt in water
    C) Water clarity
    D) Amount of nitrogen

26. What is added during potable (drinking) water treatment to kill harmful bacteria?
    A) Nitrogen
    B) Fluoride
    C) Chlorine or ozone
    D) Phosphate

27. If a population of small fish decreases, the population of large fish that eat them will likely:
    A) Increase
    B) Decrease
    C) Stay the same
    D) Become extinct immediately

28. When algae die and decompose, dissolved oxygen levels:
    A) Increase sharply
    B) Decrease due to bacterial activity
    C) Stay constant
    D) Double

29. Which of these provide the most nitrates in rivers and lakes?
    A) Runoff from fertilizer
    B) Tree roots
    C) Rocks dissolving
    D) Sunlight

30. Which of the following is an example of a predator-prey relationship?
    A) Fish eating algae
    B) Algae using sunlight
    C) Frogs eating insects
    D) Decomposers breaking down waste

Short Answer (3 pts each):

1. Why are mayflies, caddisflies, and stoneflies considered good indicators of clean water?

2. If a stream has mostly leeches, midges, and flatworms, what can you infer about the water quality? Explain your reasoning.

3. How can scientists use the diversity of macroinvertebrates (rather than just the total number) to judge water quality?

4. What does the presence of aquatic sowbugs and scuds suggest about the condition of a stream?

5. Why might air-breathing insects like water striders and whirligig beetles survive even in very polluted water, while mayflies cannot?

6. Explain how changes in pollution levels over time could shift the types of macroinvertebrates found in a wetland.

TRUE OR FALSE (1 point each)

1. High water temperature increases the amount of dissolved oxygen available for aquatic organisms.

   TRUE FALSE

2. Phosphates can enter water bodies through soap, fertilizer, and wastewater runoff.

   TRUE FALSE

3. A neutral pH is 5.5 on the pH scale.

   TRUE FALSE

4. Turbidity affects how much sunlight reaches underwater plants.

TRUE FALSE

5. When algae die and decay, bacteria use oxygen, causing oxygen levels to drop.

TRUE FALSE

6. Dissolved oxygen and biochemical oxygen demand (BOD) usually increase together.

TRUE FALSE

7. Water with high total solids will always appear clear.

TRUE FALSE

8. Colder water generally holds more dissolved oxygen than warm water.

TRUE FALSE

9. Eutrophication is the process in which nutrient pollution leads to excessive algal growth and oxygen depletion.

TRUE FALSE

10. A watershed only includes rivers and lakes, not the surrounding land.

TRUE FALSE

11. Primary wastewater treatment removes most dissolved chemicals from the water.

TRUE FALSE

12. Secondary wastewater treatment uses biological processes to break down organic matter.

TRUE FALSE

13. The nitrogen cycle helps convert atmospheric nitrogen into forms usable by plants and aquatic life.

TRUE FALSE

14. An increase in turbidity can decrease photosynthesis in aquatic plants.

TRUE FALSE

15. Invasive species can alter entire food webs by reducing native populations.

TRUE FALSE

Fill in the missing spaces in this diagram of the Nitrogen Cycle on the Image Sheet. (4 points)

1. _____

2. _____

3. _____

4. _____

Table: Water sample results from Oak Creek.

Parameter	Site 1 (upstream)	Site 2 (midstream)	Site 3 (downstream)
pH	7.2	6.4	5.8
Temperature (°C)	10	14	18
Dissolved Oxygen (mg/L)	10.5	7.6	4.3
Nitrates (mg/L)	0.3	2.1	5.6
Phosphates (mg/L)	0.1	0.6	2.4
Turbidity (NTU)	2	8	20

1. (2 pt) Based on the data, which site likely has the *highest biological oxygen demand (BOD)*?

2. (2 pt) Which site shows evidence of *eutrophication*?

3. (2 pt) What relationship do you observe between *nutrient levels (nitrates/phosphates)* and *dissolved oxygen*?

4. (1 pt) Which site most likely has *the least biodiversity*?

5. (1 pt) Suggest one likely pollution source causing the downstream conditions

SPEED ID! Identify the common name of each organism. If it's a bioindicator, name its class (Pollusion Sensitive, Moderately Sensitive, etc.). If not, state it's an aquatic nuisance plant/animal. (15 points)

Write your answer corresponding to the number of each image.

1.	2.	3.
4.	5.	6.
7.	8.	9.
10.	11.	12.
13.	14.	15.

Salinometer Testing (10 points) (Tiebreaker)

DO NOT WRITE ANYTHING HERE UNTIL YOU ARE TOLD TO

Proposed Solution Percent:

Milpitas Invitational - Repaired practice edition

Division B | Water Quality | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: A branching attached organism is photographed. Can shape distinguish coral from sponge?

Answer: _____
_____
_____